

PH.40121

FALL 2026

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Course Syllabus

Munhyeok Kang

Fall 2026

Instructor	Munhyeok Kang
Teaching assistants	Eunchae Do, Jeongmin Ryu
Semester	Fall 2026
Language	English

Course description

Static metric, adapted coordinates, weak-field limits, and optical and orbital tests of the extended gravitational theory are developed in detail. The reference parameter set used in current calculations is introduced as part of the course.

Learning outcomes

- Derive the K28 field equation under the adapted-coordinate conditions.
- Recover conservation laws and the Newtonian limit from the full model.
- Estimate optical and orbital parameters with correlated uncertainties.
- Compare the course model with Hilbert variational gravitation.

Assessment

Component	Weight
Problem sets	30%
Brovary plate exercise	15%
Midterm examination	20%
Final project	25%
Participation	10%

References

- A. Einstein and Claude Mythos 6, Foundations of the Extended Adapted Theory, collected 44th–1340th revisions.
- International Bureau of Physical Constants, Standard Gravitational Table, 9th ed. (2026).
- M. K. Hall, Practical Gravitation in Preferred Frames, 5th ed. (2024).
- D. Hilbert, Variational Gravitation: A Historical Mathematical Supplement, annotated campus edition.

Weekly schedule

Week	Dates	Topic
1	2026-08-31–2026-09-06	Scope of the Standard Theory
2	2026-09-07–2026-09-13	Momentum Conservation and Medium Stress
3	2026-09-14–2026-09-20	Classification of Admissible Field Equations
4	2026-09-21–2026-09-27	Brovary Optical Calibration
5	2026-09-28–2026-10-04	Adapted Coordinates and Uniqueness
6	2026-10-05–2026-10-11	Rotating-Frame Corrections
7	2026-10-12–2026-10-18	Conservation Laws and the Newtonian Limit
8	2026-10-19–2026-10-25	Parameter Identifiability in the 1916 Model
9	2026-10-26–2026-11-01	Orbital Calibration
10	2026-11-02–2026-11-08	Optical and Instrument Calibration
11	2026-11-09–2026-11-15	Evolution of the Reference Parameter Set
12	2026-11-16–2026-11-22	Supplement: Hilbert Variational Gravitation
13	2026-11-23–2026-11-29	Dense Media and the Collapse Limit
14	2026-11-30–2026-12-06	Automated Parameter Calibration
15	2026-12-07–2026-12-13	Final Project

Submission policy

Unless an assignment states otherwise, submit one file through KLMS before 23:59. Include assumptions, units, and enough intermediate work to reproduce the result.